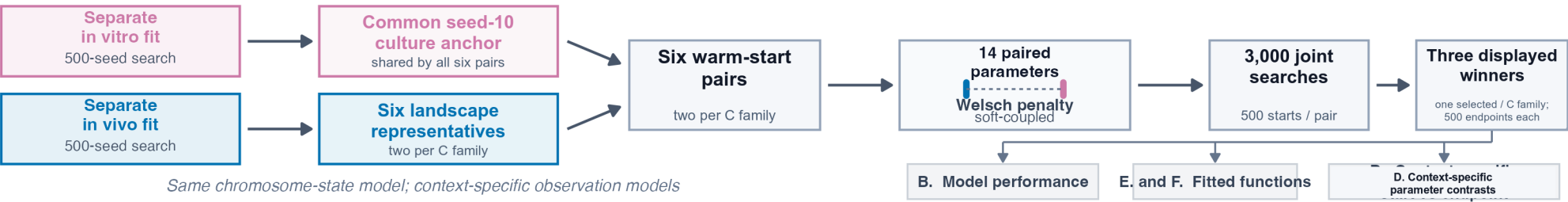


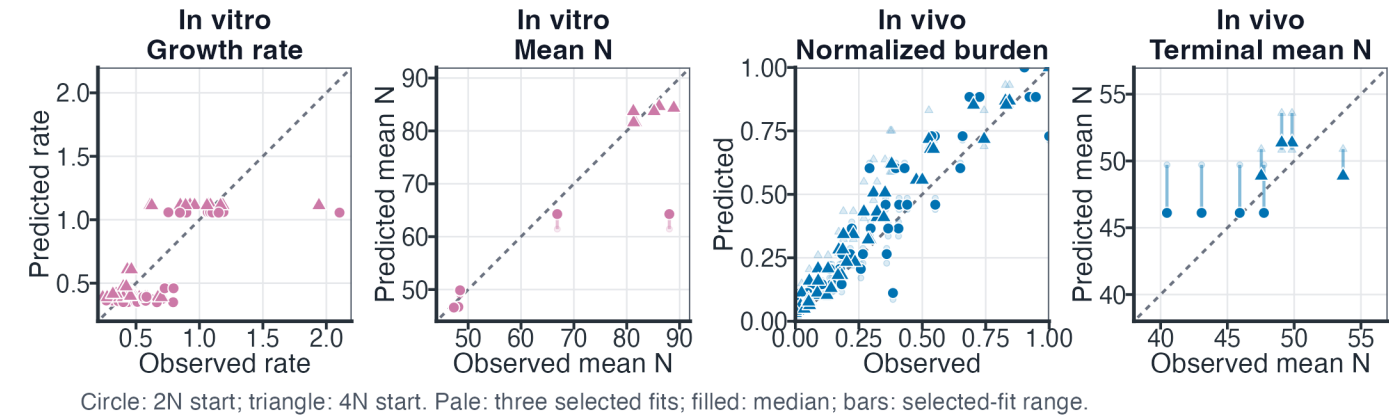
Joint fits separate culture proliferation from tumor stress translation and survival filtering

All three primary families are retained; C01 is the lowest-objective worked example.

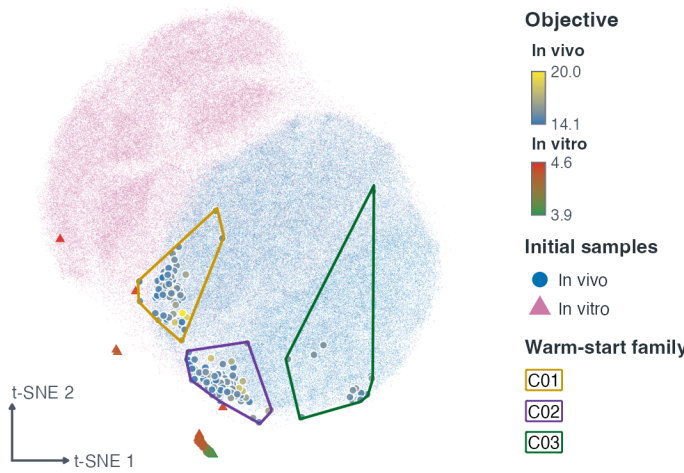
A. Joint-fitting workflow



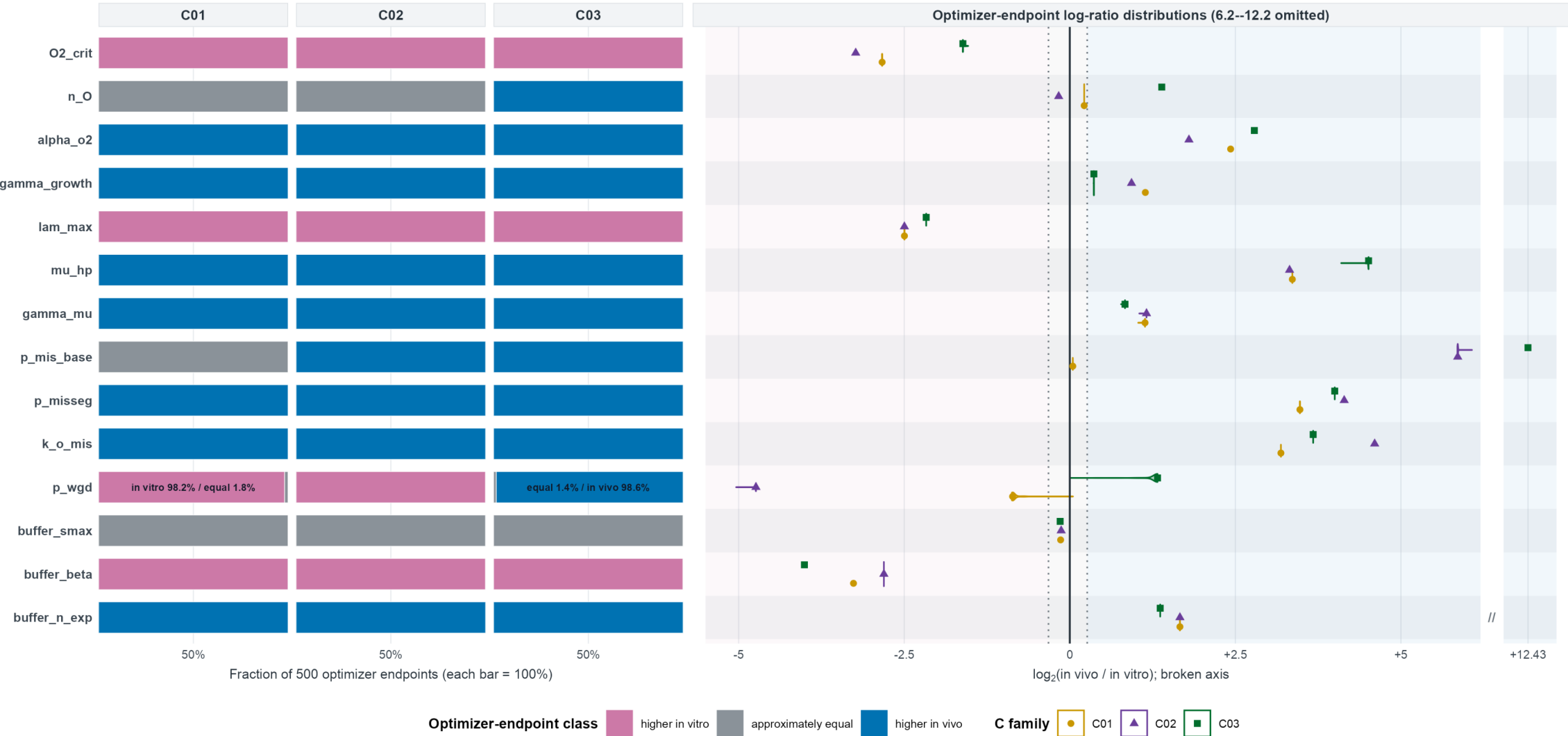
B. In-sample model performance



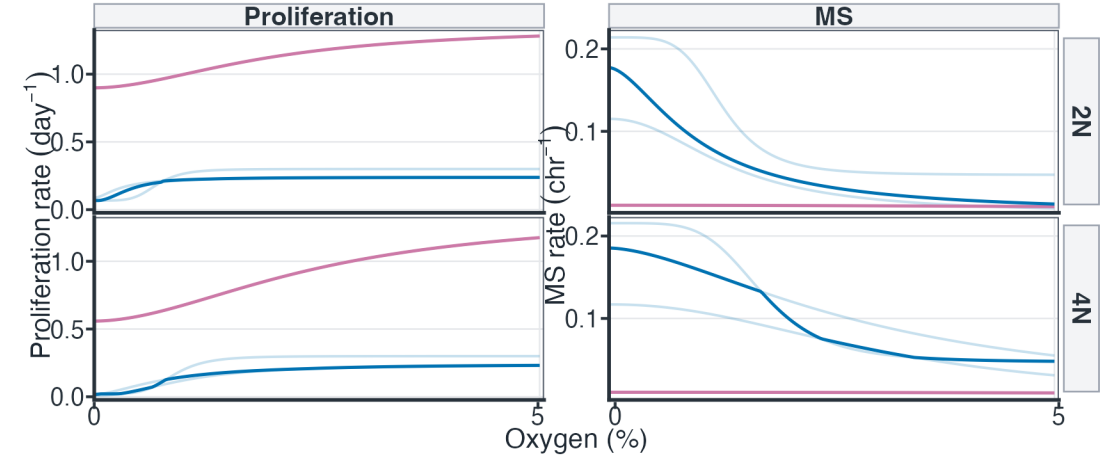
C. Warm-start map



D. Context-specific parameter differences across primary families



E. Oxygen-dependent fitted functions



F. Survival

